

PALAK SHAH

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SUMMARY

Accomplished **Data Scientist** with extensive training in **machine learning** and **big data analytics**, holding a Master's degree in Data Science and a background in Operations Management. Skilled in extracting actionable insights from vast datasets using advanced analytics tools such as **Python**, **SQL**, **Tableau**, and **PowerBI**. Demonstrates strong problem-solving capabilities and a keen ability to forecast trends and behaviors through **statistical modeling** and **predictive analytics**. Eager to apply rigorous **data mining** techniques and **data visualization** expertise to drive business solutions and innovation in a dynamic data science environment.

EDUCATION

MASTER OF SCIENCE IN DATA SCIENCE

April 2023

Simplilearn, Online

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Visualization, Statistical Methods for Decision Making, Big Data Technologies, Python Programming, Advanced Analytics, Business Intelligence

POST GRADUATE DEGREE IN OPERATIONS MANAGEMENT

July 2021

Thakur Institute of Management Studies and Research, Mumbai, India

Relevant Coursework: Supply Chain Management and Logistics, Total Quality Management (TQM), Service Operations Management

BACHELOR OF COMMERCE (B. COM)

July 2018

Kes Shroff College of Arts and Commerce, Mumbai, India

WORK EXPERIENCE

OPTICOM DATA PRODUCTS

Mumbai, India

Business Development Manager

February 2023 to March 2024

- Strategized and implemented effective business development initiatives, leading to a 15% increase in client engagement through targeted data analytics.
- Developed a predictive model that successfully forecasted market trends, aiding in a pivotal investment decision that enhanced company profitability by 10%.
- Utilized machine learning techniques to analyze business, market, and competitor data, resulting in enhanced strategic decisions and operational efficiencies.

RELIANCE RETAIL LTD.

Mumbai, India

Departmental Manager

July 2021 to August 2022

- Transformed a financially underperforming department from loss to profit within a year by optimizing inventory management and implementing effective merchandising strategies.
- Led a team of 20+ employees, fostering a culture of safety and efficiency that decreased shrinkage by 30% and increased department sales goals by 20%.
- Reviewed and analyzed sales and inventory data, identifying trends that contributed to a 25% improvement in customer satisfaction and loyalty.

PROJECTS

CAPSTONE REAL ESTATE MACHINE LEARNING PROJECT

Simplilearn, Data Science Master's Degree Program

Jan 2023 - Mar 2023

- **Objective:** Provide a banking institution with predictive insights into mortgage-backed securities and real estate investment opportunities across various U.S. regions.
- **Techniques Used:** Data engineering, exploratory data analysis, factor analysis to ensure normal distribution of predictive variables.
- **Tools & Technologies:** Python, Matplotlib, Seaboard, Tableau.
- **Outcome:** Created a statistical model predicting potential loan demands with an accuracy of 98.8% R² score. Developed a Tableau dashboard refreshing periodically for dynamic data visualization, significantly enhancing decision-making processes.

HEALTHCARE DIABETES PREDICTION MODEL

- **Objective:** Predict diabetes presence among patients using diagnostic measurements for the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK).
- **Techniques Used:** Descriptive and inferential statistics, data imputation for missing values, and machine learning model comparison (Random Forest vs. KNN).
- **Tools & Technologies:** Python, Matplotlib, Seaborn, Plotly, Scikit-learn.
- **Outcome:** Achieved robust model performance with detailed analysis via sensitivity, specificity, and ROC curves. Created an interactive Tableau dashboard displaying key metrics and trends to assist healthcare professionals.

MERCEDES BENZ MACHINE LEARNING PROJECT

- **Objective:** Implement an algorithmic solution to reduce the testing time of customized Mercedes-Benz cars, thereby increasing production line efficiency and lowering emissions.
- **Techniques Used:** Data cleansing by removing variables with zero variance, handling null values, applying label encoding, and performing dimensionality reduction.
- **Tools & Technologies:** Python, XGBoost, Pandas, NumPy.
- **Outcome:** Developed models that significantly decreased the time cars spend on the test bench without compromising safety standards, contributing to a more sustainable production process.

ATLIQ HOSPITALITY POWER BI PROJECT

- **Objective:** Develop a comprehensive Power BI dashboard for Atliq, a fictional hotel chain, to simplify the analysis of key performance metrics.
- **Techniques Used:** Data cleaning using MySQL, data reshaping with Power Query, metric calculation using DAX.
- **Tools & Technologies:** MySQL, Power BI.
- **Outcome:** Delivered a dynamic dashboard that provides weekly and monthly performance insights, enabling quick and informed decision-making for hotel management.

CORE COMPETENCIES

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|---------------------------|----------------------------|
| • Statistical Analysis | • Critical Thinking |
| • Predictive Modeling | • Project Management |
| • Machine Learning | • Strategic Planning |
| • Data Visualization | • Collaboration |
| • Big Data Handling | • Team Leadership |
| • Business Intelligence | • Communication Skills |
| • Programming Proficiency | • Presentation Skills |
| • Database Management | • Continuous Learning |
| • Problem Solving | • Professional Development |

TECHNICAL SKILLS

- **Programming Languages:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, R (basic)
- **Data Visualization Tools:** Tableau, PowerBI, Matplotlib, Seaborn
- **Machine Learning:** Regression, Classification, Clustering, Time Series Analysis, Neural Networks, Decision Trees
- **Big Data Technologies:** Hadoop, Spark (basic understanding)
- **Database Management:** MySQL, PostgreSQL
- **Statistical Analysis:** A/B Testing, Hypothesis Testing
- **Data Manipulation:** Data Cleaning, Data Wrangling, Data Integration
- **Version Control:** Git, GitHub
- **Cloud Platforms:** AWS (Amazon Web Services) Basics, Microsoft Azure Fundamentals